

Metadata Extraction (CPP-009)

CPP-Identifier	CPP-009
CPP-Label	Metadata Extraction
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Change history	Comments
Version 1.0.0 - 2025-08-29	Milestone version
Version 1.1.0 - 2025-10-16	Migration to XML
Version 1.2.0 - 2026-03-09	Fixed input of step 5

1. Description of the CPP

The TDA extracts characteristics (such as size, image dimensions, video codec, audio run time, creating application).

Inputs and outputs

Input(s)			
Data	File or, in some specific cases, Representation		
Metadata	Technical Metadata (Format identifier)		
Documentation/ guidance	Policies and detection methods for significant properties, risk properties and quality properties		
	Metadata recording Policy		
Output(s)			
Metadata	Technical Metadata		
	Provenance metadata		
	Optional	Descriptive Metadata	
		Structural Metadata	
		Rights Metadata	
		Errors and Warnings	

Definition and scope

Metadata extraction is the process of analysing a File or a set of Files (i.e. a PREMIS Representation) by means of metadata extractor tools in order to retrieve its characteristics in an automated way. In the digital preservation community, this operation is sometimes referred to as “Characterisation”. Within EDEN, however, characterisation is used to identify all operations aiming to extract properties from digital Files (CPP-008 File Format Identification, CPP-009 Metadata Extraction, and CPP-010 File Format Validation).

Metadata Extraction is generally performed at a File level; in some cases it has to be applied to a complex file structure that is not wrapped in a container File. For example, moving images stored as a sequence of DPX Files are handled as a whole by the metadata extractor tool MedialInfo.

File properties as gathered through Metadata Extraction are generally considered “Technical metadata”. However, by parsing the File, the process also extracts a wide range of its internal Descriptive, Provenance and Rights metadata.

Knowledge of these characteristics is a key requirement for many subsequent operations. In particular, it is essential for CPPs producing new Representations (i.e. CPP-026 File Normalisation, CPP-027 File Repair, CPP-014 Format Migration, and CPP-028 Creation of Derivatives) as they require further Metadata beyond the file format information as provided

by CPP-008. Audiovisual Files are the most obvious example: Most identification tools provide information about the only container format, while any of the operations mentioned above will need at least the video and audio stream format. This is equally true for all other data types: TIFF may contain image streams compressed by different algorithms, PDF 1.7 might be portfolios containing arbitrary Files, etc.

There are different types of extraction tools available: a) Generalist metadata extractor tools which are able to perform metadata extraction on a great variety of file formats of different content types (e.g. [Exiftool](#)), b) content-specific tools which cover most of the file formats for a specific content type (e.g. [MedialInfo](#) for AV Files), c) format-specific tools which are specialised on a particular format (e.g. [EPUBcheck](#) for EPUBs, [metaflac](#) for FLAC Files, etc.). Metadata extraction therefore relies on a format identification and is performed differently depending on the file format. TDAs might decide to apply several metadata extractor tools if a single tool cannot extract all required properties.

Metadata extraction - like any parsing operation - can fail and result in diagnostic error messages. These errors require systematic analysis to inform troubleshooting. In particular, the following issues may be detected by Metadata extraction errors:

- Encrypted Files;
- Truncated or broken Files
- Files assigned incorrect file format information.

The output of the metadata extractor tools should be recorded in the Information Package, as Technical, Descriptive, Rights or Provenance Metadata. It may be recorded directly as-is, or mapped to a metadata standard according to the TDA policy on metadata recording.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
Ingest	CPP-029 (Ingest)
Re-run of metadata extraction because of the release of a new metadata extractor tool or tool version	
Verify the output of processes creating new Files or Representations	CPP-014 (File Migration), CPP-026 (File Normalisation), CPP-027 (File Repair), CPP-028 (Creation of Derivatives)

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓↑ sequence					
1	CPP-008 (File Format Identification)	Format identifier	Select a suitable metadata extractor tool for the File(s), depending on its format identifier and on requirements from Significant Properties Definition, Data Quality Assessment and Risk Properties Definition and Extraction	Metadata extractor tool If relevant: configure the tool settings - syntax (XML, JSON, CSV, etc.), format (e.g., EBUCore, PBCore, in case of an AV File) and verbosity level.	
	CPP-022 (Significant Properties Definition)	Significant properties policy and detection method			
	CPP-023 (Risk Properties Definition and Extraction)	Risk properties policy and detection method			
	CPP-019 (Data Quality Assessment)	Quality properties policy and detection method			
2		File		Tool(s) output	

No	Supplier	Input	Steps	Output	Customer
		Metadata extractor tool configured according to selected settings	Applying one or sometimes several metadata extractor tool(s).	<i>Optional:</i> Errors and warnings	
3		Errors and warnings	<i>Optional:</i> Analyse the errors and troubleshoot, e.g., by removing encryption		
4		Tool(s) output	<i>Optional:</i> Map the extractor tool(s) output to metadata standard(s) The output of the extractor tool may be recorded as-is in the Information package, or may be mapped into a technical metadata standard (e.g., MIX for still images).	Tool output in standard format(s)	
		Policy on metadata recording			
5		Tool output (raw or in a standard format)	Recording the results in the Information Package (i.e. in practice, in the digital archive database and/or in the physical Information Package)	Technical metadata, optionally other types of metadata, recorded in the Information Package	CPP-029 (Ingest)
6			Document the event and its datetime	Provenance metadata	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
All preservation actions beyond bit-level preservation are based on a comprehensive understanding of the File's characteristics.	Files of poor quality may be unidentified. Derivatives may be unadapted to the end users' needs. The result of a migration may be partial, as some parts of the source File may not have been identified, thus not been copied to the target File.
Metadata extraction involves accessing the contents of the File. Hence it is an essential means to detect problematic Files (including errors like: corrupted, non-conformant to format specification, encrypted or password protected, wrong file format identification etc.).	Problematic Files are not detected.

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
/	/	/

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
Required by	CPP-002	Checksum Validation	CPP-002 relies on fixity information as produced and stored by CPP-001, when triggered by CPP-025 Enabling Access and CPP-006 AIP Batch Export. When triggered by CPP-029 Ingest CPP-002 rather relies on the fixity information supplied in the SIP.
Required by	CPP-008	File Format Identification	The selection of an appropriate extractor tool depends on file format information.
Required by	CPP-019	Data Quality Assessment	The selection of an appropriate extractor tool depends on requirements from the Data Quality Assessment CPP.
Required by	CPP-022	Significant Properties Definition	The selection of an appropriate extractor tool depends on requirements from the Significant Properties Definition CPP.
Required by	CPP-023	Risk Properties Definition and Extraction	The selection of an appropriate extractor tool depends on requirements from the Risk Properties Definition and Extraction CPP.
Required by	CPP-012	Risk Mitigation	Preservation actions (i.e. migration, emulation) in the storage depend on the identification of Files that share the same properties.
Required by	CPP-014	File Migration	File format identification is generally limited to an indication of the container format, while migration can apply to any property of the Files. Technical metadata extraction is required to both assess the compliance of files format to the Archive's format policy and control the outcome of the migration.
Required by	CPP-016	Metadata Ingest and Management	Any metadata that was extracted from the File needs to be stored, searchable and retrievable
Required by	CPP-019	Data Quality Assessment	Metadata extraction returns Metadata that are used to assess the File quality (e.g. for an audiovisual File quality assessment may rely on Metadata such as bit depth, sampling frequency, etc.).
Required by	CPP-023	Risk Properties Definition and Extraction	Metadata extraction returns Metadata that are used to identify preservation threats (e.g. for a PDF, the presence of an open password).

Relation	CPP-ID	CPP-Title	Relationship description
Required by	CPP-024	Enabling Discovery	Some Metadata provided to the consumer must have been extracted from the Files.
Required by	CPP-029	Ingest	Metadata extraction is one of the core processes that must be performed as part of Ingest.
May be required by	CPP-010	File Format Validation	Depending on the precision of the format registry used in the format identification process, the resulting information may be insufficient for selecting the right validation tool. In such cases, additional Metadata from an extraction tool may be required. For example, if an organisation uses Unix File as its identification tool, which does not distinguish between different PDF “flavours”, and wants to validate PDF/A against the PDF/A standard. In that case, metadata extraction will be necessary to identify the conformance level and select veraPDF as the suitable validation tool
May be required by	CPP-021	AIP Versioning	The documented event, datetime, and Provenance metadata from the metadata extraction may be required by AIP Versioning.
May be required by	CPP-027	File Repair	Tools extracting properties of the File or Representation are useful (and sometimes even necessary) for identifying erroneous format structures.
Affected by	CPP-018	Community Watch	Either due to changing significant properties or due to updated tools, metadata extraction requirements can be affected.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	Quality control checks	Section R10 (Quality assurance) implicitly requires metadata extraction as one of the processes implementing “quality control checks in place [that] ensure the completeness and understandability of data and metadata”.
Nestor Seal	Technical metadata collect[ion]	C30 Technical metadata
ISO 16363		/

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS		/
PREMIS	Metadata extraction Characterization	The PREMIS Data Dictionary mentions this operation as “characterization” (p. 249, section Special Topics / Format information), but the event type vocabulary maintained by the PREMIS Editorial Committee at id.loc.gov uses the term “ metadata extraction ”.

4. Reference implementations

Use cases

Metadata Extraction from AV material

Institutional background	
Institution	Bibliothèque nationale de France, FR
Description	
Trigger event	AV material must be analysed by proper tools, beyond format identification, in particular because format identification generally returns information about the container, while AV Files are wrapping streams of different nature and format
Problem statement	Discussion with AV experts required that several quality properties be extracted, in particular properties related to the group of pictures . XML was the preferred syntax for the extractor tool output, as it could be easily wrapped in METS Files.
Proposed solution	BnF has selected the tool MedialInfo as its extractor tool for AV Files, according to requirements collected by BnF. MPEG-7, one of its output formats, being standardised and expressed in XML, was selected as the format for Metadata to be stored in Archival Information Packages. As MedialInfo provides natively MPEG-7 as one of its output formats, no mapping from the tool output to a standard metadata format was required.

Publicly available documentation

Institution	Organisation type	Language	Hyperlink
TIB – Leibniz Information Centre for Science and Technology and University Library, DE	National library; Non-commercial digital preservation service; Research infrastructure; Research performing organisation	English	https://wiki.tib.eu/confluence/spaces/lza/pages/93608618/Ingest

Institution	Organisation type	Language	Hyperlink
CSC – IT Center for Science Ltd., FI	Non-commercial digital preservation service	English	https://urn.fi/urn:nbn:fi-fe2020100578096 (section 5)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/preservation/preservation-planning/#characterization